DPCN 2026 · GROUP 24 · STATISTICS REPORT

Research output of three CSIR chemistry labs, 2016–2026

CSIR-NCL (Pune) · CSIR-IICT (Hyderabad) · CSIR-CECRI (Karaikudi) · OpenAlex, collected 13 Oct 2026 (2026 partial)

Institute	Papers	Journal	Conference	Author rows	Citations	Q1 (SJR)	Roster scientists	With ≥1 paper
CSIR-IICT	6,167	6,086	81	20,183	125,231	55.1%	168	129
CSIR-NCL	5,404	5,353	51	14,726	134,679	67.4%	166	151
CSIR-CECRI	2,885	2,830	55	7,484	93,125	73.1%	51	26
All three (distinct)	14,352	14,166	186	42,393	350,920	63.3%	385	306

How the data was built

- OpenAlex works of the three ROR-verified institutes; articles, reviews, letters and conference papers
- Faculty = scientific cadre on official staff pages, matched to OpenAlex authors in the corpus
- Quartile = SCImago (SJR) best quartile of the publication year (2025 values for 2026)

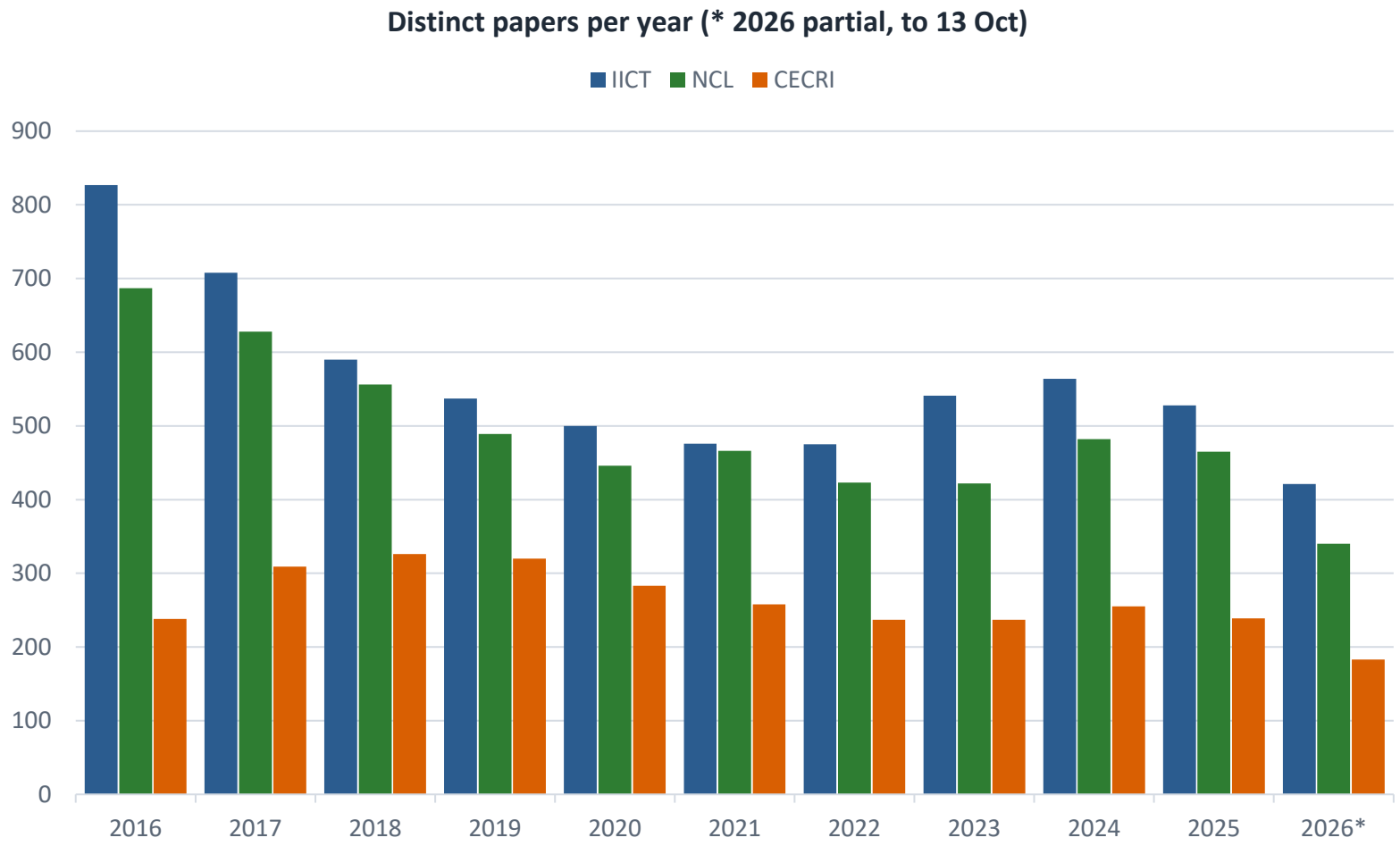
Limits

- OpenAlex author and affiliation matching is imperfect; papers may be missing or mis-attributed
- h-index uses 2016–2026 corpus papers only; students and project staff are not faculty
- SJR may differ from JCR; none of the 186 conference papers is CORE-ranked

Interactive dashboard: https://prathm01s.github.io/dpcn_almost_stable/

Publication trend

14,352 distinct papers in 2016–2026; IICT 43% of them, NCL 38%, CECRI 20%



Totals and growth

IICT: 6,167 papers;
CAGR 2016→2025 −4.9%; peak year 2016

NCL: 5,404 papers;
CAGR 2016→2025 −4.2%; peak year 2016

CECRI: 2,885 papers;
CAGR 2016→2025 +0.1%; peak year 2018

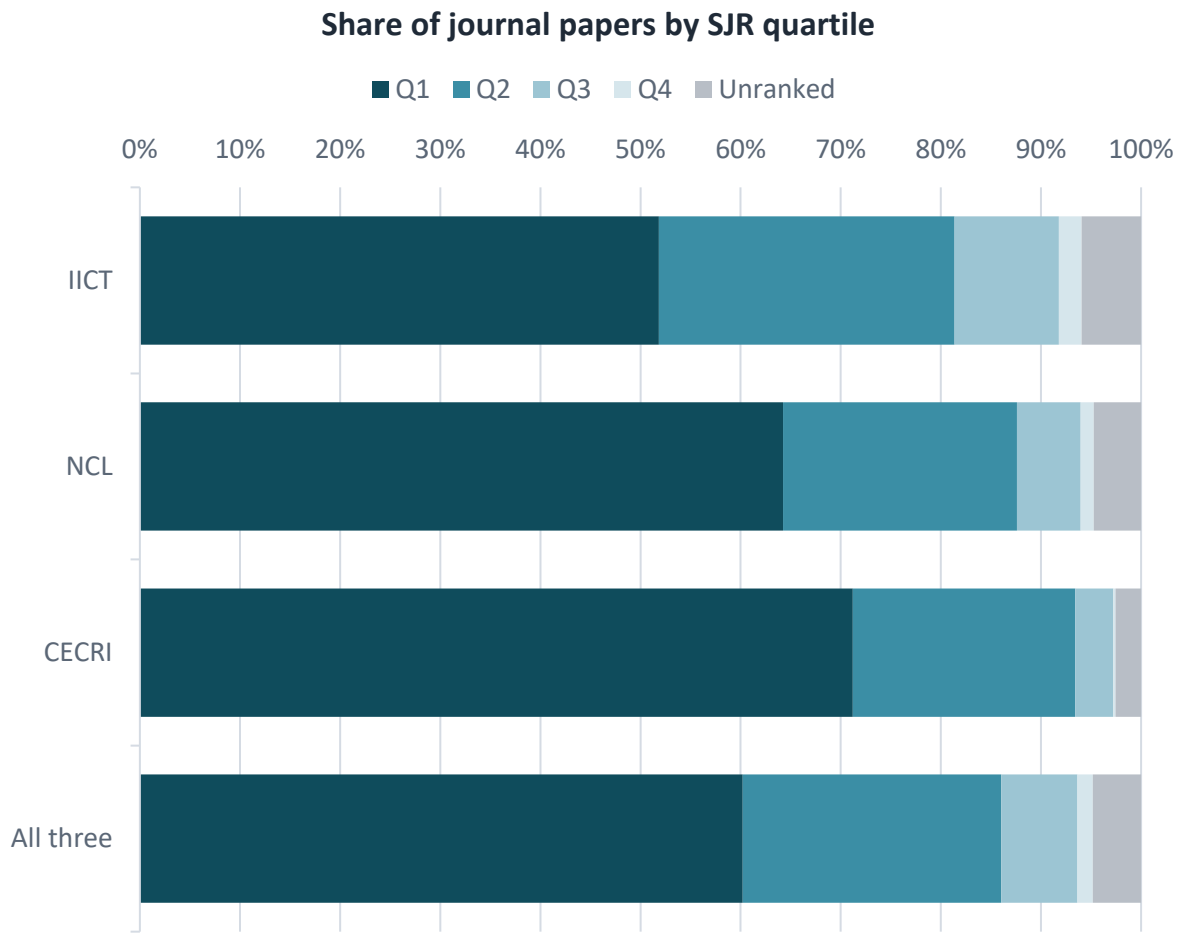
Observations

- Combined output went from 1,739 papers in 2016 to 1,220 in 2025
- Growth rates differ in sign across the three labs
- 2026 covers papers to 13 Oct only (934 so far) and is left out of CAGR

Source: stats/tables/publication_trend.csv · Papers counted once per institute; combined counts distinct papers

Journal quartiles and citation impact ratio

63.3% of ranked journal papers are in SJR-Q1 journals; Q1 by impact ratio (IR > 0.59, top 25% of papers) ranges from 22.0% (IICT) to 34.0% (CECRI)



Q1 by journal vs Q1 by impact ratio (% of papers)

Institute	Q1 by journal (SJR)	Q1 by IR (IR > 0.59)	Institute IR
IICT	55.1%	22.0%	0.41
NCL	67.4%	24.1%	0.47
CECRI	73.1%	34.0%	0.59
All three	63.3%	25.2%	0.47

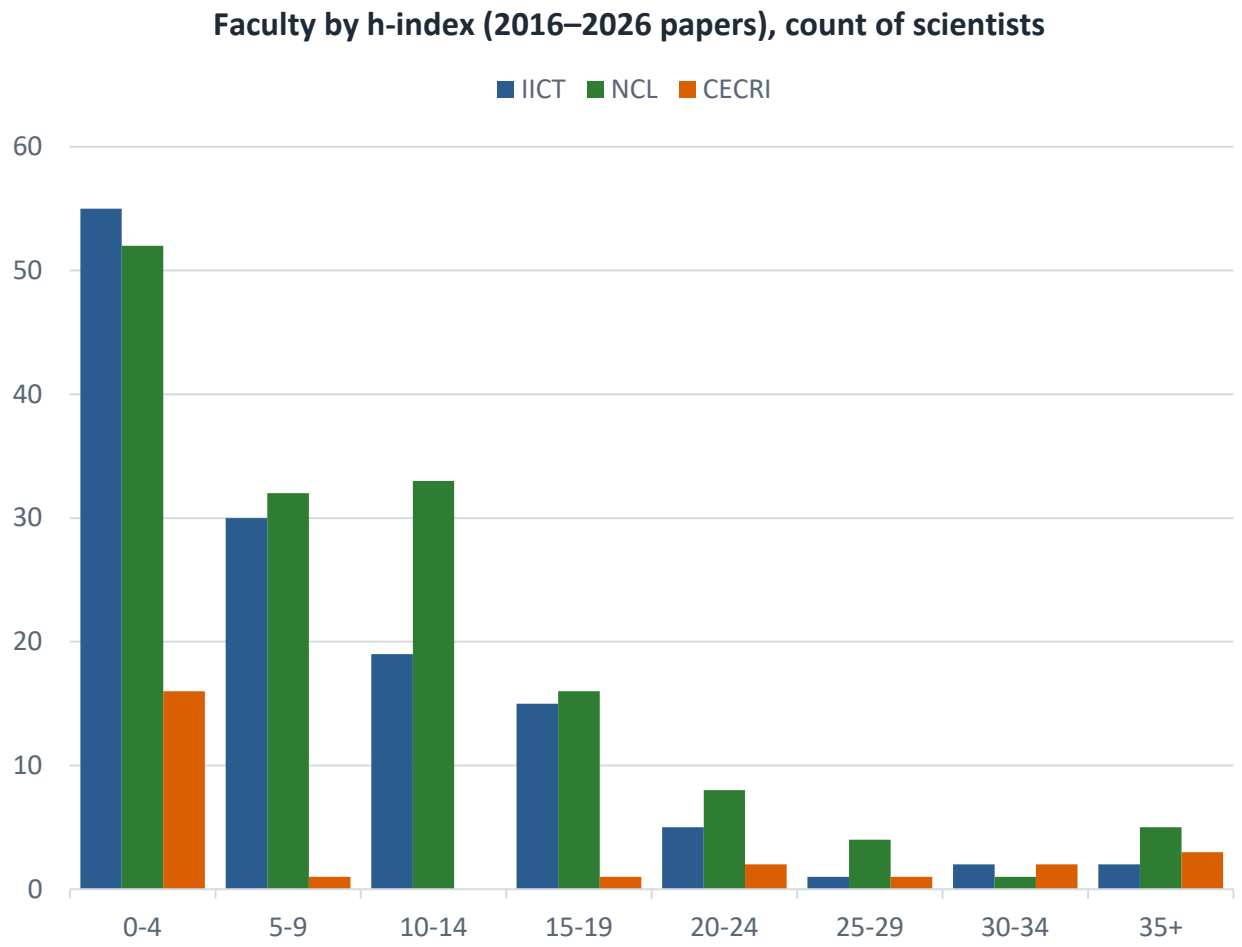
Impact ratio (IR) = citations received ÷ citations made (references). A paper is IR-Q1 when IR exceeds the threshold; the institute IR is all citations received ÷ all references made.

- Threshold IR > 0.59 is the 75th percentile of IR over all papers, so IR-Q1 is the top 25% of papers — the same share a quartile covers (median paper IR 0.26).
- The two definitions disagree: only 32.9% of papers in SJR-Q1 journals are top-25% by IR. SJR ranks the journal; IR ranks the paper.
- Older papers have had longer to be cited; 513 papers with no references in OpenAlex are excluded.

Source: stats/tables/quartile_distribution.csv, impact_ratio_summary.csv, impact_ratio_vs_sjr.csv · SJR best quartile (Scopus-based); IR from OpenAlex citations (13 Oct) and references

Faculty h-index distribution

Median h-index of the 306 roster scientists with papers is 7; CECRI 2.5, NCL 8, IICT 6



Institute	Faculty	Median	Mean	P75	Max
IICT	129	6	8.1	12	54
NCL	151	8	9.8	14	44
CECRI	26	2.5	11.9	23.5	59
All	306	7	9.3	14	59

Highest h-index (2016–2026 window)

Top faculty by h-index	Lab	Papers	Citations	h
Dr Subrata Kundu	CECRI	260	15,767	59
Dr. S. Venkata Mohan	IICT	230	11,389	54
Dr. Sreekumar Kurungot	NCL	181	8,978	44
Dr. Subashchandrabose Chi…	NCL	57	3,617	41
Dr. Ujjwal Pal	IICT	148	4,758	40
Dr. Vinod C Prabhakaran	NCL	152	4,752	39

Source: stats/tables/h_index_summary.csv, h_index_distribution.csv, faculty_performance.csv · h-index on 2016–2026 corpus papers

Top 5 journals per institute

Most frequent venues: IICT – ChemistrySelect (239); NCL – ChemistrySelect (140); CECRI – Electrochimica Acta (77)

CSIR-IICT 6,167 papers

#	Journal	Papers	Share	Q
1	ChemistrySelect	239	3.9%	Q2
2	Organic & Biomolecular Chemistry	144	2.4%	Q1
3	Tetrahedron Letters	128	2.1%	Q2
4	ChemInform	122	2.0%	—
5	The Journal of Organic Chemistry	120	2.0%	Q1

CSIR-NCL 5,404 papers

#	Journal	Papers	Share	Q
1	ChemistrySelect	140	2.6%	Q2
2	Chemical Communications	116	2.2%	Q1
3	The Journal of Organic Chemistry	106	2.0%	Q1
4	ACS Omega	97	1.8%	Q1
5	Organic Letters	94	1.8%	Q1

CSIR-CECRI 2,885 papers

#	Journal	Papers	Share	Q
1	Electrochimica Acta	77	2.7%	Q1
2	Journal of Materials Chemistry A	68	2.4%	Q1
3	ChemistrySelect	67	2.4%	Q2
4	New Journal of Chemistry	64	2.3%	Q1
5	International Journal of Hydrogen…	59	2.1%	Q1

All three combined: ChemistrySelect (443); New Journal of Chemistry (270); RSC Advances (245); Chemical Communications (239); The Journal of Organic Chemistry (225)

Source: stats/tables/top_journals.csv · ranked by distinct papers; share = % of the institute's papers; Q = SJR quartile

Top 10 most-cited papers (all three institutes)

350,920 citations in total; mean 24.4 and median 11 per paper; the top paper has 2,493

#	Title	Journal	Year	Lab	Citations	DOI
1	Recent Trends and Perspectives in Electrochemical Water Splitting with an Emphasis on Sulfi…	ACS Catalysis	2016	CECRI	2,493	10.1021/acscatal.6b02479
2	Cancer chemotherapy and beyond: Current status, drug candidates, associated risks and progr…	Genes & Diseases	2022	IICT	1,826	10.1016/j.gendis.2022.02.007
3	Precision and correctness in the evaluation of electrocatalytic water splitting: revisiting…	Energy & Environmental Science	2018	CECRI	1,623	10.1039/c7ee03457a
4	Covalent Organic Frameworks: Chemistry beyond the Structure	Journal of the American Chemi…	2018	NCL	1,574	10.1021/jacs.8b10334
5	Phytochemicals in Cancer Treatment: From Preclinical Studies to Clinical Practice	Frontiers in Pharmacology	2020	NCL	1,085	10.3389/fphar.2019.01614
6	Selective Molecular Separation by Interfacially Crystallized Covalent Organic Framework Thi…	Journal of the American Chemi…	2017	NCL	1,060	10.1021/jacs.7b06640
7	Cooperativity in Noncovalent Interactions	Chemical Reviews	2016	IICT	932	10.1021/cr500344e
8	“The Fe Effect”: A review unveiling the critical roles of Fe in enhancing OER activity of N…	Nano Energy	2020	CECRI	931	10.1016/j.nanoen.2020.105514
9	Functional Naphthalene Diimides: Synthesis, Properties, and Applications	Chemical Reviews	2016	IICT	927	10.1021/acs.chemrev.6b00160
10	Polylactic acid: synthesis and biomedical applications	Journal of Applied Microbiolo…	2019	NCL	873	10.1111/jam.14290

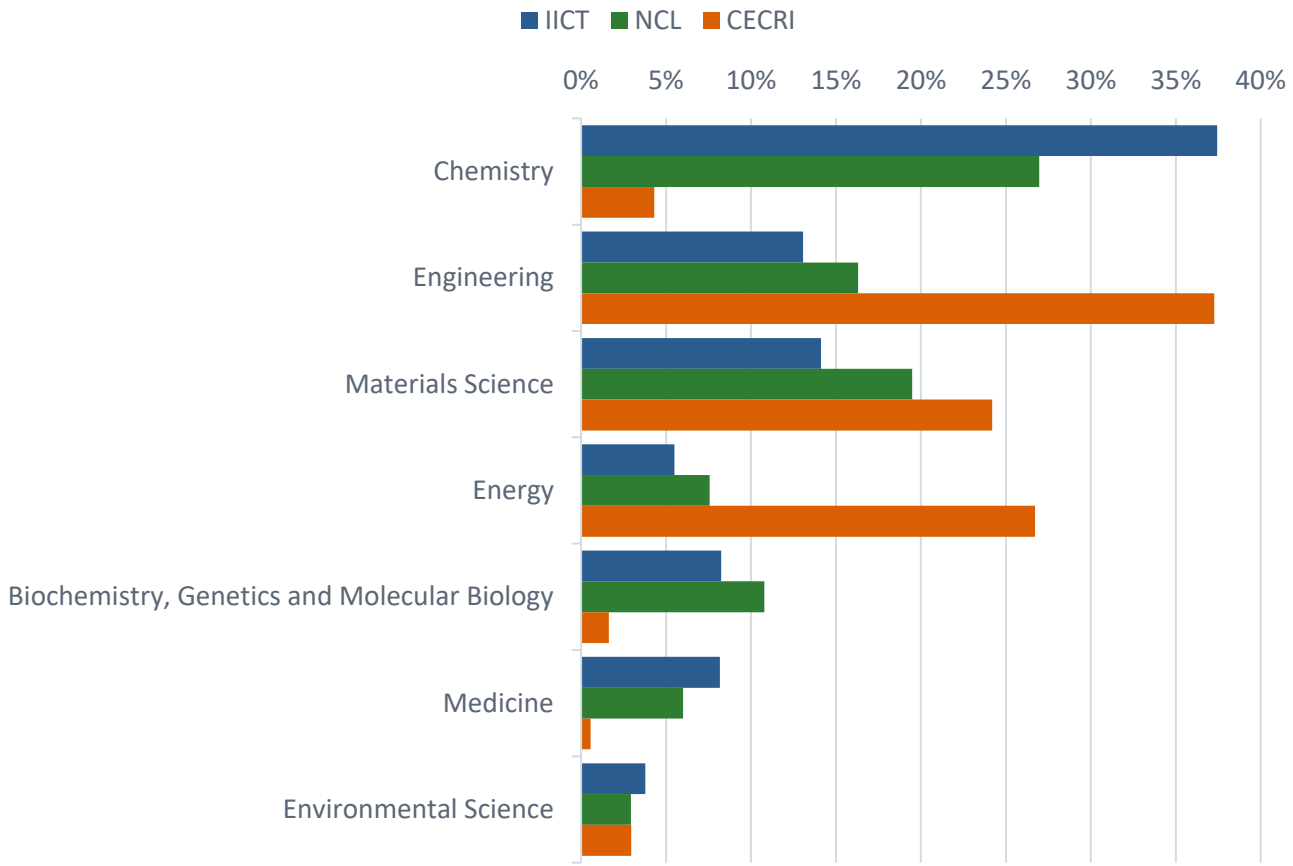
IICT: 125,231 citations, mean 20.3, uncited 12.2% · NCL: 134,679 citations, mean 24.9, uncited 11.0% · CECRI: 93,125 citations, mean 32.3, uncited 6.8%

Source: stats/tables/top_cited_papers.csv, citation_summary.csv · OpenAlex cited_by_count on 13 Oct 2026; older papers have had longer to accrue citations

Research areas and departments

Largest field: IICT – Chemistry (37%), NCL – Chemistry (27%), CECRI – Engineering (37%)

Share of each institute's papers by OpenAlex field



Top 2 departments per institute by papers

Department	Lab	Faculty	Papers	Q1
Polymers & Functional Materials	IICT	15	498	68%
Applied Biology	IICT	16	477	56%
Physical and Materials Chemistry	NCL	26	1,195	73%
Organic Chemistry	NCL	37	764	59%
Electrochemical Process Engineering	CECRI	6	350	89%
Electrodics and Electrocatalysis	CECRI	2	142	75%

Department = roster division of each faculty member; a paper counts in every department of its faculty authors. Papers by non-roster authors only are not assigned to a department.

Source: stats/tables/research_areas_field.csv, department_performance.csv · field = OpenAlex primary topic → field

Citation network

27,268 citation links among 10,988 corpus papers; 1,653 cross between labs. With other Indian institutions: 88,639 citations in, 87,292 out

10,988
papers in the in-corpus citation graph

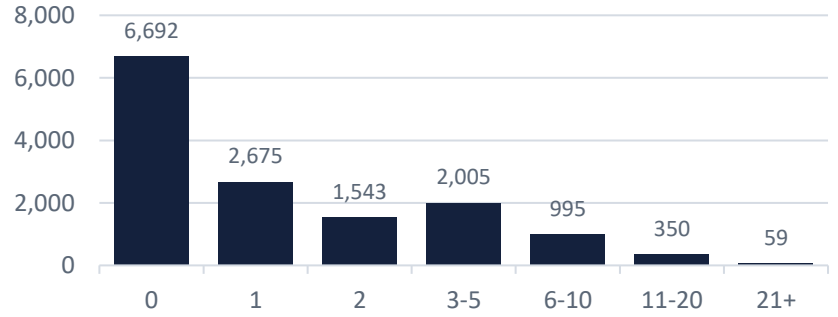
7,627
papers cited ≥1 time within corpus (max in-degree 129)

1,653
citation links between the three labs

1.01
in / out citation ratio with other Indian institutions

#	PageRank top 10	Lab	Year	In-deg	PR
1	Recent Trends and Perspectives in Electrochemical Water…	CECRI	2016	129	0.0030
2	Pt Nanoparticle Anchored Molecular Self-Assemblies of D…	CECRI	2016	41	0.0028
3	Unprotected and interconnected Ru 0 nano-chain networks…	CECRI	2016	30	0.0027
4	Functional Naphthalene Diimides: Synthesis, Properties,…	IICT	2016	53	0.0020
5	Precision and correctness in the evaluation of electroc…	CECRI	2018	110	0.0018
6	Waste biorefinery models towards sustainable circular b…	IICT	2016	71	0.0017
7	Regulation of acidogenic metabolism towards enhanced sh…	IICT	2016	40	0.0016
8	Pharmaceutical cocrystals: walking the talk	NCL	2016	36	0.0016
9	Facile and scalable route to sheets-on-sheet mesoporous…	CECRI	2016	18	0.0015
10	Recent advances in N-heterocyclic carbene (NHC)-catalys…	NCL	2016	28	0.0015

In-degree distribution (corpus papers)



Citation flow with other Indian institutions

Lab	Cited by Indian	Cites Indian	In/out	Citing insts
IICT	32,840	38,891	0.84	1,816
NCL	32,434	30,206	1.07	1,683
CECRI	23,365	18,195	1.28	1,310

Source: stats/tables/network_summary.csv, network_citation_*.csv, citation_flow_summary.csv · in-degree = citations from other corpus papers; flows count links to DOI-bearing papers

Authorship network

7,865 authors, 35,426 co-author pairs; Louvain finds 131 communities (modularity 0.82), almost all within a single lab

7,865
authors; 96% in the largest connected component

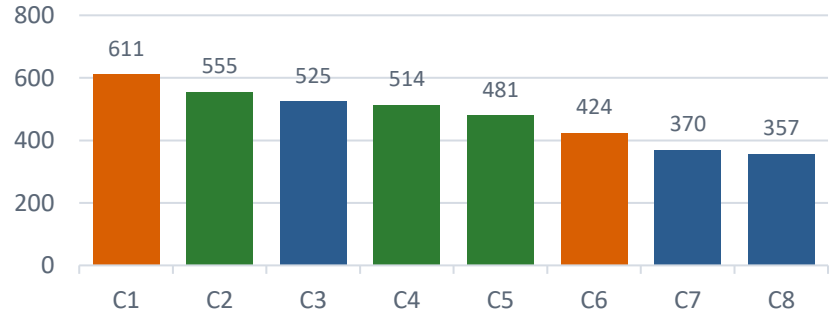
9.0
mean number of distinct co-authors

131
Louvain communities (32 with ≥10 authors)

525
co-authorship links between the three labs

#	Most collaborative (degree)	Lab	Degree	Betweenness	Btw rank	Comm.
1	Balasubramanian Sridhar	IICT	380	0.098	#1	3
2	Dr. Jagadeesh Babu Nanubolu	IICT	285	0.097	#2	16
3	Dr. Rajesh G. Gonnade	NCL	280	0.059	#5	5
4	Dr. Sunil Misra	IICT	198	0.056	#7	9
5	Dr. B.V. Subba Reddy	IICT	196	0.057	#6	3
6	Dr. Kumar Vanka	NCL	195	0.045	#9	5
7	Dr. Sreekumar Kurungot	NCL	154	0.025	—	4
8	Chityal Ganesh Kumar	IICT	152	0.012	—	8
9	Dr. Andugulapati Sai Balagi	IICT	151	0.017	—	10
10	Ramakrishna Sistla	IICT	150	0.065	#4	7

Largest communities (colour = main lab)



Top brokers by betweenness

Balasubramanian Sridhar IICT · 0.098 · degree 380

Dr. Jagadeesh Babu Nanubolu IICT · 0.097 · degree 285

Dr Subrata Kundu CECRI · 0.068 · degree 137

Ramakrishna Sistla IICT · 0.065 · degree 150

Top Indian partners: Academy of Scientific and Innovative Research (641); Savitribai Phule Pune University (375); Council of Scientific and Industrial Research (337)

Source: stats/tables/network_summary.csv, network_authorship_*.csv, indian_partner_institutions.csv · interactive graphs: https://prathm01s.github.io/dpcn_almost_stable/

Ten non-obvious findings

Each finding answers one question; charts and method notes are in the dashboard's Insights tab

#	Finding	Result
1	Friendship paradox	93.2% of researchers have fewer co-authors than their co-authors do. The average researcher has 9.2 co-authors, but the average co-author has 31.1.
2	Rich club	The top 1% most-connected researchers show no rich-club effect and are less interconnected than chance (ρ_norm = 0.75). 39 of 78 are CSIR faculty.
3	Network fragility	Removing the top 841 bridge researchers (11.2% of the network) halves the cluster; random removal needs 3,282. Removing 644 of them separates the three labs' faculty into different components.
4	Concentration (Price's law)	√N = 18 scientists produce 35.8% of the cluster's papers (Price's law predicts 50%). The top 10% produce 48.2%; Gini = 0.61 (papers), 0.74 (citations).
5	Journal quartile vs impact	11.3% of the cluster's top-10% most-cited papers were published outside Q1 journals, while 1.4% of Q1 papers are still uncited.
6	Local vs global influence	In-cluster influence and global citations correlate only moderately (Spearman ρ = 0.40). 43 papers are 'hidden gems': top-5% internal influence, below-median global citations.
7	Sleeping beauties	2 of the 7 strongest candidates (top 1% of 602 papers by Ke's B) show sustained delayed recognition. The strongest, 'Funnel shaped molecules containing benzo/pyrido[1,2,5]thiadiazole f…' (IICT, 2016), drew 1.4 citations/yr for 5 years, then 4.7/yr from 2021.
8	COVID shock	Output was 1.8% above the 2016–19 trend in 2020 and 12.7% above in 2021. COVID-related work peaked at 5.5% of CSIR-IICT's papers in 2021; the topic mix shifted about as much as in a normal period (JSD ratio 1.05).
9	Brain circulation	17.0% of classifiable CSIR scientists are returnees (NCL 17.7% (n=141), IICT 17.0% (n=106), CECRI 11.1% (n=18)); their abroad spell was most often in the US (25 of 45). Their papers' citations do not differ significantly from home-grown peers of similar career age (+8%, 95% CI -20% to +46%); 20.6% of returnees' papers vs 16.9% of home-grown scientists' papers have a foreign co-author.
10	Succession risk	In CECRI, 46.3% of recent papers (48.6% of their citations) have only roster authors likely to retire within ~5 years (IICT 20.8%, NCL 11.5%). Highest exposure: Electrochemical Process Engineering, CECRI (74.6% of 193 papers).

Source: stats/tables/insights/*.json (generated by src/stats/insights) · charts: https://prathm01s.github.io/dpcn_almost_stable/